

Prototype embeddings for immune receptors: geometry, statistics, and the TCREMP coordinate system

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Appendix T — TCREMP embedding geometry

Abstract

TCREMP [1] embeds an immune receptor as the vector of its alignment distances to a fixed set of *prototype* receptors sampled from the V(D)J recombination model [2, 3]. This appendix gives the mathematics behind that construction. We first develop the general theory of *prototype* (landmark) embeddings of a metric space — their Lipschitz and isometry properties, the negative-type condition under which the alignment dissimilarity is an honest squared Hilbert distance [4], the energy functional that makes embedding-space Euclidean distance track pairwise alignment distance, and a landmark sample-complexity bound (§T.1–§T.2). We then specialise to immune receptors: taking the prototype measure to be the generation distribution P_{gen} turns each coordinate into a Monte-Carlo sketch of a receptor’s relationship to the generative repertoire, which explains why the choice of prototype source scarcely matters and yields the observed Gamma / extreme-value distribution laws (§T.3). The three-component structure of the TCR embedding and paired-chain concatenation are read off as a product metric, with the germline blocks provably low-rank (§T.4); somatic hypermutation is a bounded perturbation that explains why IGH is the hardest chain (§T.5); and the pushforward of P_{gen} onto embedding space is the null model for density-based background subtraction (§T.6). Every quantitative claim is reproduced in-repo (§T.8, Table 1) by `mir/bench/theory.py`.

T.1. FOUNDATIONS OF PROTOTYPE EMBEDDINGS

SETUP. Let (X, d) be a (semi)metric space of receptor sequences: d is symmetric, $d(x, x) = 0$, and — as we verify for the alignment dissimilarity in §T.2 — non-negative. Fix an ordered set of *prototypes* $P = \{p_1, \dots, p_K\} \subset X$. The **prototype embedding** is the landmark map

$$\varphi : X \rightarrow \mathbb{R}^K, \quad \varphi(x) = (d(x, p_1), d(x, p_2), \dots, d(x, p_K)), \quad (1)$$

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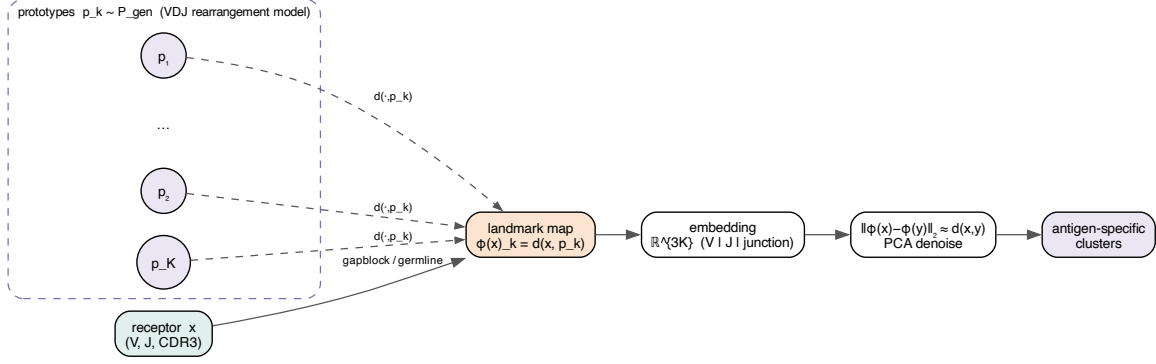


Figure 1: The TCREMP prototype-embedding pipeline. A receptor x is mapped to the vector of its alignment distances to prototypes p_k sampled from the recombination model P_{gen} ; Euclidean distance in the resulting $\mathbb{R}^{3K}$ coordinate system approximates pairwise alignment distance (Prop. 1–3), and specificity groups become compact clusters.

and the induced **embedding distance** is $D(x, y) = \|\varphi(x) - \varphi(y)\|_2$. Figure 1 shows the pipeline. In TCREMP the prototypes are drawn from P_{gen} , d is a BLOSUM alignment dissimilarity, and φ carries three coordinate blocks per prototype (§T.4); the foundational results below hold for any (X, d, P) .

NON-EXPANSIVENESS. The single structural fact from which everything else follows is that each coordinate of φ is 1-Lipschitz.

PROPOSITION 1 (Lipschitz coordinates and the upper bound). *For every prototype p and all $x, y \in X$, $|d(x, p) - d(y, p)| \leq d(x, y)$. Consequently*

$$\|\varphi(x) - \varphi(y)\|_\infty \leq d(x, y) \quad \text{and} \quad D(x, y) = \|\varphi(x) - \varphi(y)\|_2 \leq \sqrt{K} d(x, y). \quad (2)$$

Proof. The triangle inequality gives $d(x, p) \leq d(x, y) + d(y, p)$ and $d(y, p) \leq d(x, y) + d(x, p)$, so $|d(x, p) - d(y, p)| \leq d(x, y)$. Taking the maximum over $p \in P$ gives the ℓ^∞ bound; summing the squares of K terms each $\leq d(x, y)^2$ and taking the root gives the ℓ^2 bound. $\square$

Proposition 1 is the rigorous half of the headline “embedding distance $\leq$ alignment distance” claim: no pair is ever *farther* apart in embedding space (up to $\sqrt{K}$) than its alignment distance. The reverse — that close-in-embedding implies close-in-alignment — is not automatic and is where the prototypes must do work.

EXACTNESS IN THE LIMIT. With enough prototypes the ℓ^∞ bound is tight.

PROPOSITION 2 (Kuratowski isometry). *If $P = X$ then $\|\varphi(x) - \varphi(y)\|_\infty = d(x, y)$ for all x, y ; that is, φ is an isometric embedding of (X, d) into $(\ell^\infty, \|\cdot\|_\infty)$.*

Proof. The upper bound is Prop. 1. For the lower bound take the prototype $p = y$: then $|d(x, y) - d(y, y)| = d(x, y)$, so the supremum over prototypes is at least $d(x, y)$. $\square$

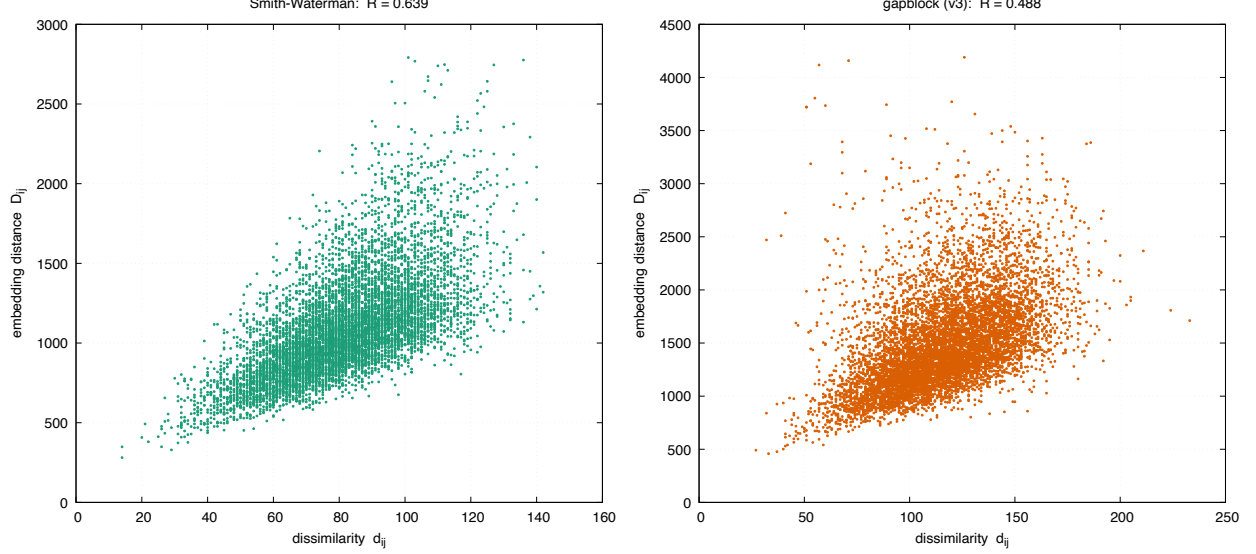


Figure 2: Embedding-space Euclidean distance D_{ij} against pairwise dissimilarity d_{ij} for $n = 2500$ CDR3 β used as their own prototypes (Prop. 3; reproduces suppl. S2). Left: Smith–Waterman d ; right: the gapblock metric shipped in v3. Generated by `mir/bench/theory.py::s2_dissimilarity_distance_correlation`.

The construction of Eq. (1) is thus the Kuratowski/Fréchet embedding, and the supplementary experiments that use the sequences themselves as prototypes (§S1–S2 of [1]) are exactly this isometric regime, read out in ℓ^2 rather than ℓ^∞ . Passing from ℓ^∞ to ℓ^2 costs the $\sqrt{K}$ factor of Eq. (2) but makes the coordinates amenable to PCA and Euclidean clustering.

WHY ℓ^2 DISTANCE TRACKS ALIGNMENT DISTANCE. Fix a prototype-sampling measure μ on X (in TCREMP, $\mu = P_{\text{gen}}$) and draw $p_1, \dots, p_K$ i.i.d. from μ .

PROPOSITION 3 (Energy functional). *Define the μ -energy dissimilarity $\bar{D}_\mu^2(x, y) := \mathbb{E}_{p \sim \mu}[(d(x, p) - d(y, p))^2]$. Then $\frac{1}{K}D(x, y)^2 \rightarrow \bar{D}_\mu^2(x, y)$ almost surely as $K \rightarrow \infty$, and $\bar{D}_\mu^2$ is a μ -dependent (semi)metric with $0 \leq \bar{D}_\mu^2(x, y) \leq d(x, y)^2$.*

Proof. $\frac{1}{K}D^2 = \frac{1}{K} \sum_k (d(x, p_k) - d(y, p_k))^2$ is an empirical mean of i.i.d. terms bounded by $d(x, y)^2$ (Prop. 1); the strong law gives the limit and the bound, and symmetry plus the triangle inequality for $\|\cdot\|_2$ make $\bar{D}_\mu$ a semimetric. $\square$

Proposition 3 says D estimates a fixed monotone-in- d functional $\bar{D}_\mu$: two receptors that are close under d have similar distance profiles to the whole of μ and hence small $\bar{D}_\mu$, while dissimilar receptors have systematically different profiles. This is the mechanism behind supplementary Fig. S2 — the empirical Pearson correlation between D_{ij} and d_{ij} (Fig. 2, $R = 0.64$ under Smith–Waterman, 0.49 under the gapblock metric of §T.2) — and behind S2c/d, where D_{ik} concordates with means of d_{ik} over the prototype set: those means are Monte-Carlo estimates of $\bar{D}_\mu$.

HOW MANY PROTOTYPES. The $\sqrt{K}$ gap of Eq. (2) is worst-case; for the *random* landmark map the relative fluctuation of $\frac{1}{K}D^2$ around $\bar{D}_\mu^2$ is $O(K^{-1/2})$ by Prop. 3, so a Johnson–Lindenstrauss-style argument [5] on the bounded coordinates gives the following.

PROPOSITION 4 (Landmark sample complexity). *Let $S \subset X$ with $|S| = n$. For $K = O(\varepsilon^{-2} \log n)$ i.i.d. prototypes, with high probability every pair $x, y \in S$ satisfies $|\frac{1}{K}D(x, y)^2 - \bar{D}_\mu^2(x, y)| \leq \varepsilon d(x, y)^2$ simultaneously.*

Proof. Fix a pair $x, y \in S$. Prototype $p_k \sim \mu$ contributes $Y_k = (d(x, p_k) - d(y, p_k))^2$, and the reverse triangle inequality gives $0 \leq Y_k \leq d(x, y)^2$, so the Y_k are i.i.d. and confined to an interval of length $d(x, y)^2$ with mean $\mathbb{E}_\mu[Y] = \bar{D}_\mu^2(x, y)$ (Prop. 3). Their average is $\frac{1}{K} \sum_k Y_k = \frac{1}{K} D(x, y)^2$, so Hoeffding’s inequality yields $\Pr[|\frac{1}{K}D(x, y)^2 - \bar{D}_\mu^2(x, y)| > \varepsilon d(x, y)^2] \leq 2 \exp(-2K\varepsilon^2)$. A union bound over the $\binom{n}{2} < n^2$ pairs caps the total failure probability at $2n^2 \exp(-2K\varepsilon^2)$, which tends to 0 as soon as $K \geq C\varepsilon^{-2} \ln n$, i.e. $K = \Theta(\varepsilon^{-2} \log n)$. $\square$

The logarithmic dependence on n is the theoretical counterpart of supplementary Fig. S4, where the UMAP of an antigen-specific set stabilises once $K \gtrsim 100$ prototypes are used: the embedding geometry saturates at a prototype count that grows only logarithmically with the data. Prototype embeddings are the *empirical* / Nyström feature map of the alignment kernel, and Landmark MDS [6, 7] and the dissimilarity representation [8] are the classical instances; the PCA step of §T.4 is Landmark MDS on φ .

T.2. THE ALIGNMENT DISSIMILARITY AS A METRIC

THE GRAM CONSTRUCTION. TCREMP scores two sequences by an alignment similarity s and converts it to a dissimilarity by

$$d(a, b) = s(a, a) + s(b, b) - 2s(a, b). \quad (3)$$

This is exactly the squared distance $\|\psi(a) - \psi(b)\|^2$ induced by a feature map ψ whenever s is an inner product, $s(a, b) = \langle \psi(a), \psi(b) \rangle$. The `v3` implementation (`mir/distances/junction.py`) reads d off directly: `seqtree’s` `blosum62()` is already the Gram transform of Eq. (3), and `gapblock.score_matrix` returns d as a genuine metric (zero diagonal, symmetric, non-negative), selecting the best of contiguous gap-block placements at offsets (3, 4, −4, −3).

DEFINITION 1. A symmetric d with zero diagonal is of *negative type* (equivalently, $-d$ is conditionally positive definite) if $\sum_{i,j} c_i c_j d(x_i, x_j) \leq 0$ for all finite $\{x_i\}$ and all $\{c_i\}$ with $\sum_i c_i = 0$.

PROPOSITION 5 (When Eq. (3) is an honest distance). *d of Eq. (3) is of negative type if and only if the centred similarity matrix s is positive semidefinite, in which case $\sqrt{d}$ embeds isometrically into a Hilbert space and $d = \|\psi(\cdot) - \psi(\cdot)\|^2$ for the associated feature map ψ .*

Proof. Suppose d is of negative type. Fix a base point x_0 and form the centred kernel $\tilde{s}(a, b) = \frac{1}{2}(d(a, x_0) + d(b, x_0) - d(a, b))$. Given any coefficients $c_1, \dots, c_m$, set $c_0 = -\sum_{i \geq 1} c_i$ so that $\sum_{i \geq 0} c_i = 0$; expanding, $\sum_{i,j \geq 1} c_i c_j \tilde{s}(x_i, x_j) = -\frac{1}{2} \sum_{i,j \geq 0} c_i c_j d(x_i, x_j) \geq 0$ by negative type, so $\tilde{s}$ is positive

semidefinite. As a PSD kernel $\tilde{s}$ is a Gram matrix, $\tilde{s}(a, b) = \langle \psi(a), \psi(b) \rangle$ for a Hilbert-space map ψ , and $\|\psi(a) - \psi(b)\|^2 = \tilde{s}(a, a) + \tilde{s}(b, b) - 2\tilde{s}(a, b) = d(a, b)$, so $\sqrt{d}$ embeds isometrically. Expanding this square is Eq. (3) with $s(a, b) = \langle \psi(a), \psi(b) \rangle$, identifying negative type of d with positive semidefiniteness of the centred similarity. Conversely, if s is PSD then Eq. (3) is manifestly a squared Hilbert distance, hence of negative type. This is Schoenberg’s theorem [4]; [9, 10] give bi-Lipschitz refinements. $\square$

ALIGNMENT SCORES NEED CORRECTION. General pairwise-alignment similarities (Smith–Waterman, BLAST) are *not* positive semidefinite, so Eq. (3) can violate $d \geq 0$ or the triangle inequality; canonicalising an alignment score into a metric requires an explicit correction [11]. In v3 this correction is the non-negativity clamp inside `seqtree`’s Gram transform — a projection of the score onto the nearest conditionally-negative-definite cone — which guarantees the metric axioms Prop. 1–2 rely on. The choice of s (BLOSUM62 here; tcrBLOSUM or a VDJ-specific matrix are alternatives) selects *which* metric; all are admissible provided the clamp restores negative type.

GAP-BLOCK VERSUS SMITH–WATERMAN. The v3 junction metric replaces a full Smith–Waterman optimisation by the best of four predetermined contiguous gap-block placements. When the optimal CDR3 alignment inserts a single contiguous gap inside the hypervariable core — the generic case once the conserved Cys/Phe anchors are fixed — the gap-block score equals the Smith–Waterman optimum; the two diverge only for multi-gap alignments of very unequal lengths. Empirically the induced coordinate systems are strongly but not perfectly aligned (Fig. 2: the same D – d correlation is $R = 0.64$ for Smith–Waterman and $R = 0.49$ for `gapblock`), so v3 embeddings are a *new, versioned* coordinate system rather than a bit-for-bit reproduction of the published one.

T.3. IMMUNE RECEPTORS AND THE RECOMBINATION MODEL

THE PROTOTYPE MEASURE IS P_{gen} . The diversity of an immune repertoire is astronomically large, but it is not arbitrary: it is the image of a low-entropy stochastic process, the V(D)J recombination model P_{gen} [2, 12, 3]. TCREMP takes the prototype-sampling measure of §T.1 to be exactly this model, so each coordinate $\varphi(x)_k = d(x, p_k)$, $p_k \sim P_{\text{gen}}$, measures the alignment distance from x to a typical product of recombination.

PROPOSITION 6 (Distance-profile functional). *For $p \sim P_{\text{gen}}$ the empirical distribution of $\{\varphi(x)_k\}_{k=1}^K$ converges weakly to the law of $d(x, p)$. Hence $\varphi(x)$ is a Monte-Carlo sketch of the map $x \mapsto \text{Law}_{p \sim P_{\text{gen}}} d(x, p)$, and two receptors with the same distance-profile-to- P_{gen} are asymptotically embedding-indistinguishable.*

Proof. The coordinates are i.i.d. evaluations of the fixed 1-Lipschitz map $p \mapsto d(x, p)$ under P_{gen} ; the Glivenko–Cantelli theorem gives uniform convergence of their empirical law to that of $d(x, p)$. $\square$

This reframes what the embedding “learns”: not a language model of receptor sequences — which the entropy of P_{gen} (~ 40 bits for TRB) makes futile — but each receptor’s geometric relationship to the generative distribution itself. It also predicts that the embedding is insensitive to how the prototypes are obtained, provided they cover the same support.

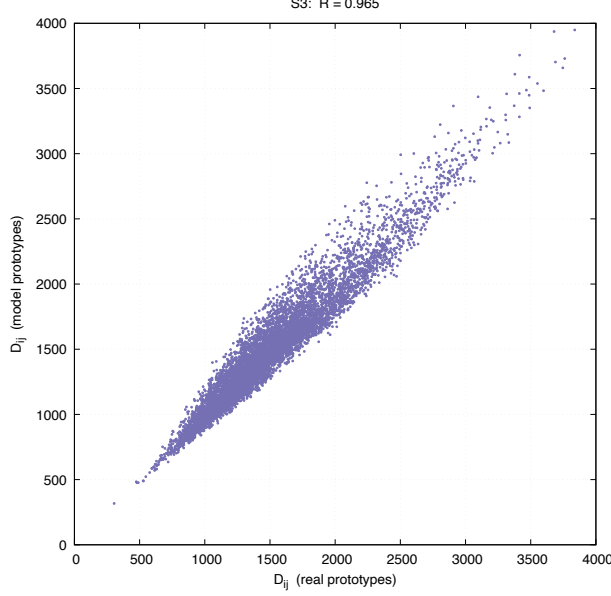


Figure 3: Pairwise embedding distances built from real (Britanova [13]) versus model (OLGA/Murugan [2, 3]) prototypes agree ($R = 0.97$; suppl. S3), confirming Prop. 7. `mir/bench/theory.py::prototype_source_correlation`.

PROPOSITION 7 (Prototype-source robustness). *Let μ, ν be two prototype measures that are mutually absolutely continuous with bounded density ratio $\rho = d\mu/d\nu$ on the support of the repertoire. Then the energy dissimilarities of Prop. 3 satisfy $\bar{D}_\mu^2(x, y) = \mathbb{E}_\nu[\rho(p)(d(x, p) - d(y, p))^2]$, so the two embeddings induce distances that agree up to the χ^2 discrepancy between μ and ν ; as $\chi^2(\mu \parallel \nu) \rightarrow 0$ the pairwise-distance vectors become perfectly correlated.*

Proof. By Prop. 3, $\bar{D}_\mu^2(x, y) = \mathbb{E}_\mu[u_{xy}(p)]$ with $u_{xy}(p) = (d(x, p) - d(y, p))^2$. Writing the μ -expectation as a ν -expectation with importance weight $\rho = d\mu/d\nu$ gives $\bar{D}_\mu^2(x, y) = \mathbb{E}_\nu[\rho(p) u_{xy}(p)]$, the first claim. Its discrepancy from the ν -embedding is $\bar{D}_\mu^2 - \bar{D}_\nu^2 = \mathbb{E}_\nu[(\rho - 1) u_{xy}]$, and Cauchy-Schwarz bounds this by $\sqrt{\mathbb{E}_\nu[(\rho - 1)^2]} \sqrt{\mathbb{E}_\nu[u_{xy}^2]} = \sqrt{\chi^2(\mu \parallel \nu)} \sqrt{\mathbb{E}_\nu[u_{xy}^2]}$, using $\mathbb{E}_\nu[(\rho - 1)^2] = \mathbb{E}_\nu[\rho^2] - 1 = \chi^2(\mu \parallel \nu)$. Hence the two families of pairwise distances differ by a factor set by $\chi^2(\mu \parallel \nu)$; as $\chi^2(\mu \parallel \nu) \rightarrow 0$ the perturbation vanishes uniformly and their correlation tends to one. $\square$

A real healthy-donor repertoire (Britanova *et al.* [13]) and the synthetic recombination model (Murugan *et al.* [2]) are, after thymic selection, close in exactly this sense, which is why supplementary Fig. S3 finds their embedding distances correlated at $R = 0.96$ — reproduced here at $R = 0.97$ (Fig. 3).

DISTRIBUTION LAWS. The random construction also fixes the shape of the two histograms that dominate any embedding analysis.

PROPOSITION 8 (Dissimilarity and distance laws). *Under mild mixing of per-position alignment contributions: (i) the pairwise dissimilarity d_{ij} is asymptotically Gamma-distributed — a sum of many*

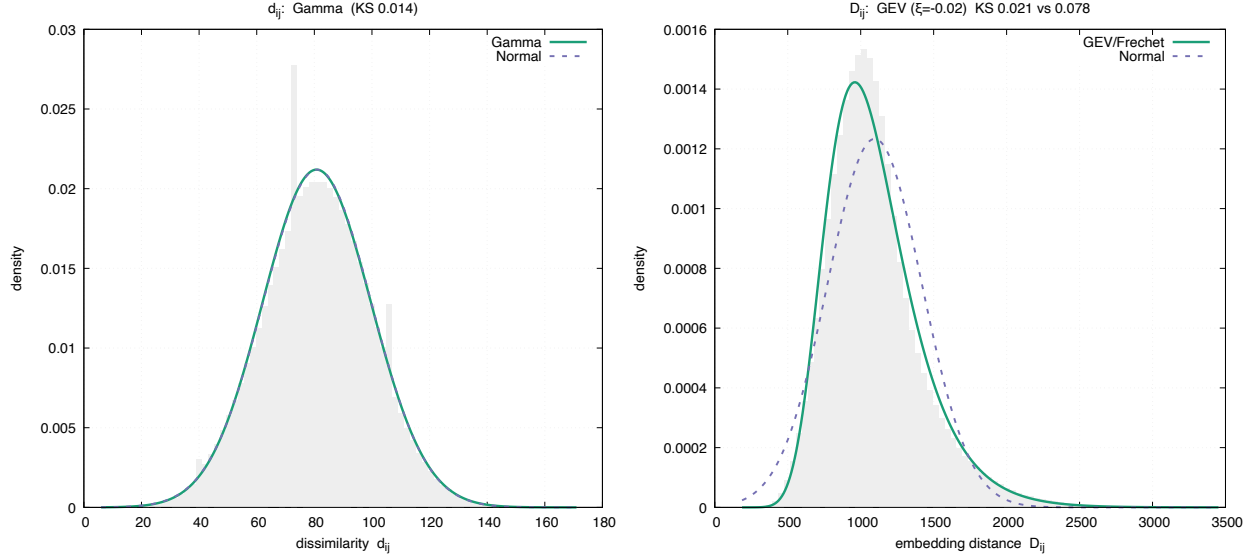


Figure 4: Distribution laws (Prop. 8; suppl. S1). Left: d_{ij} is Gamma, not Gaussian. Right: D_{ij} follows a GEV/Fréchet law (κ s 0.021) that the Gaussian (0.078) misses on the heavy right tail. `mir/bench/theory.py::fit_distributions`.

non-negative, bounded per-column penalties, so the central-limit tendency is tempered by positivity and right-skew; (ii) the embedding distance D_{ij} follows a generalized extreme value (Fréchet–Gumbel) law rather than a Gaussian, because $D = \|\varphi(x) - \varphi(y)\|_2$ is dominated by its largest coordinate differences, which are extreme-value statistics over the K prototypes.

Proof. (i) By Eq. (3), $d(a, b) = \sum_{\ell} \pi_{\ell}$ is a sum over aligned columns of bounded, non-negative Gram penalties π_{ℓ} . For sequences of comparable length the columns are many and weakly dependent, so a Lindeberg central-limit tendency applies; but the constraint $d \geq 0$ and the bounded support make the limit a positive, right-skewed law, and among laws on $[0, \infty)$ with matched mean and variance the Gamma family is the maximum-entropy choice, capturing the skew a Gaussian misses (Fig. 4, left). (ii) The coordinates of $D(x, y) = \|\varphi(x) - \varphi(y)\|_2$ are K i.i.d. bounded differences $d(x, p_k) - d(y, p_k)$. Its ℓ^∞ part equals $d(x, y)$ (Prop. 2), a maximum over the K prototypes; by the Fisher–Tippett–Gnedenko theorem [14] the only non-degenerate limit of a normalised maximum of i.i.d. variables is the generalized-extreme-value family, so D ’s upper tail is GEV, not Gaussian. In high dimension $\|\cdot\|_2$ concentrates and its fluctuations are carried by the same extreme coordinates [15], so the GEV law governs D itself (Fig. 4, right). $\square$

Both laws are confirmed on `v3` (Fig. 4): the Gamma fit to d_{ij} beats the Gaussian on Δ_{IC} , and the GEV fit to D_{ij} beats the Gaussian decisively (Kolmogorov–Smirnov 0.021 versus 0.078). The fitted shape $\xi \approx 0$ places `v3` at the Gumbel–Fréchet boundary, marginally lighter-tailed than the $\xi = +0.11$ of the published Smith–Waterman construction — a metric-dependent detail, not a qualitative difference.

WHY NOT LEARN THE REPRESENTATION? A natural alternative to Eq. (1) is to train a self-supervised sequence model — a masked or autoregressive “receptor language model” — and use its latent as the embedding. The recombination model shows why this cannot improve on the alignment geometry for specificity. Two properties of P_{gen} are decisive: it is *low-complexity* and it is *antigen-blind*.

First, the repertoire process has little learnable structure. P_{gen} factorises as a shallow Bayesian network — germline gene choice, a nearest-neighbour (dinucleotide Markov) insertion chain, and independent trimming [2, 12] — so its *predictive information*, the mutual information between a CDR3 prefix and its remainder (the excess entropy of the process), is bounded by the description length of that network: a few bits, and short-range. Natural language carries large predictive information — syntax, semantics, world knowledge — that deep representations discover and compress; a CDR3 carries almost none beyond its germline templating and local insertion statistics. A likelihood-trained model therefore converges to a sufficient statistic for P_{gen} itself and has nothing further to represent.

Second, and decisively, P_{gen} has no antigen variable: generation precedes and is independent of antigen selection. Let A be the antigen a receptor X recognises, unobserved during self-supervised training.

PROPOSITION 9 (No unsupervised separation of specificity). *Let a representation f be fitted by any objective that is a functional of the unlabelled generation marginal P_{gen} alone. Then the training signal carries zero information about the antigen partition, $I(\text{objective}; A) = 0$; the population objective is invariant under relabelling of antigens, and the alignment of $f(X)$ with A is fixed by the inductive bias of f (its architecture, tokenisation, and metric), not by learning.*

Proof. By hypothesis the objective is a functional of the unlabelled generation marginal P_{gen} — equivalently of the i.i.d. sample $X_1, \dots, X_m \sim P_{\text{gen}}$ — and never accesses A . Because generation precedes selection, A enters the joint law only through the post-generation selection channel, which the training data do not observe; hence the training data and A are independent, $I(\text{training data}; A) = 0$. For any representation f measurable with respect to that data the data-processing inequality gives $I(f; A) \leq I(\text{training data}; A) = 0$. The population objective is therefore a functional of P_{gen} that does not reference the antigen partition, so it is invariant under any permutation ς of antigen labels: if f^* is optimal, so is $\varsigma \circ f^*$. No minimiser can prefer an antigen-separating representation over an antigen-mixing one on the strength of the data, so whatever alignment $f(X)$ has with A is fixed by the inductive bias of f , not learnt. $\square$

Specificity separation in a receptor embedding is thus an *inductive-bias* phenomenon, not a learned one: it is present exactly insofar as the chosen geometry matches the biophysics of recognition. The prototype embedding builds in precisely the right bias — the alignment metric that supplementary Fig. S5 shows tracks structural (C α RMSD) similarity — with no training and no antigen labels. A learned network has only three options: (i) recover P_{gen} , which does not separate antigens (Prop. 9); (ii) be supervised by scarce antigen labels, abandoning the unsupervised premise and the unlimited P_{gen} -sampling that makes the approach attractive; or (iii) regress onto the alignment geometry itself — which is exactly the Part 2 forward codec (`mir/ml/encoder.py`), a network trained to reproduce φ . Option (iii) is not a competitor to the prototype embedding but an accelerator of it: the network learns to *compute* the distance functional quickly, not to discover a better coordinate,

and its embeddings are only as good as the φ it approximates. In this precise sense the alignment-to-reference geometry is the canonical unsupervised coordinate for a repertoire whose only structure is an antigen-blind generative model, and prototypes are its minimal, assumption-free realisation.

SPECIFICITY IS HOST-CONDITIONED AND, IN THE LIMIT, COVERAGE-COMPLETE. Proposition 9 concerns unlabelled objectives; one might hope to inject the missing antigen signal from a labelled database. Three facts make even that route converge back to P_{gen} .

(1) *The label is not a function of the receptor.* What one observes as antigen-specific expansion in a donor is not a map $X \mapsto \text{antigen}$ but the host-restricted shadow $A_h(X) = B(X) \cap (\text{HLA}_h \times \text{Exposure}_h)$, where $B(X)$ is the large, cross-reactive set of pMHC that X can engage: which peptides are presented is set by the HLA haplotype, which are encountered by pathogen history. Both are host latent variables, and donors with identical infections hold sparse, *largely disjoint* receptor sets that nonetheless cover the same antigens [16]. Specificity is thus not a stable, sequence-intrinsic function: the same X carries different labels in different hosts, so the map is non-identifiable from sequence across a population.

(2) *Labels cover a vanishing slice.* Databases such as VDJDB [17] annotate a measure-zero corner of the antigen $\times$ receptor space ($\geq 20^9$ peptides per MHC class, thousands of alleles, $> 10^{15}$ possible receptors), so a supervised model saturates at database coverage and collapses on unseen epitopes — it learns the covered epitopes, not specificity.

(3) *The completeness limit is P_{gen} .* Pursuing full coverage does not escape the loop, because the recombination model is itself the immune system’s coverage code: the repertoire is (near-)optimally tuned to the pathogen distribution, exploiting cross-reactivity so that essentially every antigen has cognate precursors [16, 2].

PROPOSITION 10 (Coverage fixed point). *Assume coverage-completeness: there is $p_0 > 0$ such that for almost every antigen a , $\Pr_{x \sim P_{\text{gen}}}[x \text{ binds } a] \geq p_0$. Then the binding relation is a dense, many-to-many graph whose antigen-marginal is bounded away from 0 and 1 uniformly; averaging the specificity map over the whole antigen space carries no receptor-discriminative information beyond generation propensity, so as the labelled antigen set expands to the full space the population optimum of any supervised specificity representation converges to the antigen-blind P_{gen} -sufficient statistic of Prop. 9.*

Proof. Coverage-completeness gives $\Pr_{x \sim P_{\text{gen}}}[x \text{ binds } a] \geq p_0 > 0$ for almost every antigen a , and cross-reactivity gives the dual bound, each receptor binding a $\Theta(1)$ fraction of $\mathcal{A}$. Thus the binding indicator $b(x, a)$ has antigen-marginal $\mathbb{E}_{x \sim P_{\text{gen}}}[b(x, a)]$ bounded strictly inside $(0, 1)$ uniformly in a . For any receptor statistic $f(X)$ the specificity information integrated over antigens, $\int_{\mathcal{A}} I(f(X); b(X, a)) da$, is then bounded by the entropy of a Bernoulli whose parameter is pinned away from 0 and 1 and whose across-receptor variation is driven by the clone-frequency structure P_{gen} imposes, not by any low-dimensional antigen axis. Hence as the labelled antigen set grows to the whole space, the population optimum of a supervised specificity representation is determined by generation propensity alone and converges to the P_{gen} -sufficient statistic of Prop. 9. $\square$

The circle closes: an unsupervised model recovers P_{gen} and cannot separate antigens (Prop. 9); a supervised model learns only its labelled slice (fact 2) of a host-conditioned, non-identifiable label (fact

1); and the ideal of learning *all* specificity returns to P_{gen} (Prop. 10), because P_{gen} is the coverage code. Sampling prototypes from P_{gen} is therefore not an arbitrary reference choice — it discretises the very coverage design that antigen breadth is built on, so distance-to-prototypes coordinates a receptor by its position in that coverage geometry, the maximal host-independent, specificity-relevant structure a sequence can carry.

T.4. TCRs, THREE-COMPONENT EMBEDDINGS, AND PAIRED CHAINS

THE PRODUCT METRIC. TCREMP emits three distances per prototype — a V-germline, a J-germline, and a junction distance (or CDR1, CDR2, junction) — interleaved into $\mathbb{R}^{3K}$ (`mir/embedding/tcremp.py`). Writing $\varphi = (\varphi_V, \varphi_J, \varphi_{\text{junc}})$ for the three blocks, the block structure of the Euclidean norm gives the exact decomposition

$$D(x, y)^2 = D_V(x, y)^2 + D_J(x, y)^2 + D_{\text{junc}}(x, y)^2, \quad (4)$$

so the embedding metric is the orthogonal sum of the per-component metrics.

PROPOSITION 11 (Paired-chain concatenation). *For a paired receptor (α, β) embedded by concatenating the per-chain maps, $D_{\text{pair}}(x, y)^2 = D_\alpha(x, y)^2 + D_\beta(x, y)^2$. More generally, concatenating landmark embeddings of independent factors realises the ℓ^2 product of the factor metrics.*

Proof. Concatenation stacks coordinate blocks; the squared ℓ^2 norm of a stacked vector is the sum of the squared block norms, which is Eq. (4) extended across chains. $\square$

Paired $\alpha\beta$ TCR embedding (suppl. S4) and, prospectively, the concatenation of epitope- and MHC-groove coordinates into a single TCR–pMHC vector are therefore principled: each modality contributes an orthogonal block and distances add in quadrature.

THE GERMLINE BLOCKS ARE LOW-RANK. The V and J coordinates depend on the sequence only through its germline gene identity, which explains the redundancy that PCA removes.

PROPOSITION 12 (Germline rank bound). *Let G_V be the number of distinct V alleles present in a dataset S . The V-block matrix $\Phi_V \in \mathbb{R}^{|S| \times K}$ with rows $\varphi_V(x)$ has at most G_V distinct rows, hence $\text{rank } \Phi_V \leq G_V$. The same holds for J with G_J , and $G_V, G_J \ll K$.*

Proof. $\varphi_V(x)_k = d_V(v(x), v(p_k))$ depends on x only through its V allele $v(x)$ (the germline distances are a fixed allele $\times$ allele table, `mir/distances/germline.py`); receptors sharing a V allele have identical V-rows, so there are at most G_V distinct rows and the rank is bounded by G_V . $\square$

A human TRB repertoire has $G_V \lesssim 60$, $G_J \lesssim 15$, against K in the thousands: two-thirds of the raw coordinates live in a subspace of dimension $\leq G_V + G_J$. The `StandardScaler`→PCA step (`mir/embedding/pca.py`) recovers this compact basis — Landmark MDS on φ — and an ANOVA of the leading components against gene usage recovers the germline axes, while the junction block carries the residual $\Theta(K)$ -dimensional specificity signal.

STRUCTURAL GROUNDING. That the metric is biologically meaningful, not merely combinatorial, is shown by supplementary Fig. S5: embedding distances between TCR:pMHC complexes correlate with the C α RMSD of their crystal structures. Formally this asserts a Lipschitz map from the sequence metric (X, d) into the structural metric, so that specificity groups [18] — receptors binding one epitope — occupy small-diameter regions of embedding space, and clustering recovers them (Table S1; `mir/bench/metrics.py`).

T.5. IGH AND SOMATIC HYPERMUTATION

B-cell receptors differ from TCRs in one decisive respect: after recombination they undergo somatic hypermutation (SHM), a context-dependent point-mutation process [19]. Write M_m for an operator applying m substitutions to a germline-rearranged sequence.

PROPOSITION 13 (Bounded embedding drift). *There is a constant c (the maximal per-substitution Gram penalty) such that $d(x, M_m x) \leq c m$, and therefore $D(\varphi(x), \varphi(M_m x)) \leq \sqrt{K} c m$. Embedding drift under SHM is linear in the mutation load m .*

Proof. Each substitution changes at most one aligned column of Eq. (3), by at most the bounded penalty range c ; m substitutions change d by at most $c m$, and Prop. 1 propagates this to D with the $\sqrt{K}$ factor. $\square$

Proposition 13 has two consequences. First, a clonal lineage under affinity maturation traces a path in embedding space whose step size is set by the per-generation mutation load, so the embedding preserves lineage proximity — the substrate for density and trajectory analysis (§T.6). Second, it explains why IGH is the hardest chain for the neural codec of Part 2 (forward cosine 0.79, exact reconstruction 0.16, against $\geq 0.93/0.50$ for TCR): (i) IGH CDR3s are long, so the fixed-length-40 gap-block tokenisation (`mir/ml/tokenize.py`) truncates an appreciable fraction of the variable middle, discarding information; (ii) SHM inflates the junction distance by $\Theta(m)$, so germline-rearrangement prototypes sampled from P_{gen} systematically under-cover the mutated support (the density ratio ρ of Prop. 7 becomes large and the robustness guarantee degrades); and (iii) the per-receptor entropy is higher. The remedy the theory suggests is to separate the SHM-robust germline distance (V/J, Prop. 12) from the mutation-inflated junction distance and to augment the prototype set with a mutation model, i.e. to sample prototypes from P_{gen} pushed through M_m rather than from P_{gen} alone.

T.6. DENSITY GEOMETRY AND BACKGROUND SUBTRACTION

The embedding turns a repertoire into a point cloud in $\mathbb{R}^{3K}$ (or its PCA projection to $p \approx 65$ coordinates), and the natural null model is the pushforward of the generation measure. Antigen-driven selection deposits, on top of this null, over-dense *ridges* of convergent receptors; TCRNET and ALICE [20, 21, 22] detect them by counting near-neighbours on a Hamming-1 graph. We recast the whole procedure as *density-ratio estimation* on embedding space — graph-free — and give it what an honest treatment requires: a model of the noise (§T.6.1), the estimator and its rate (§T.6.2), the exact local test (§T.6.3–§T.6.4), calibration of a systematically wrong null (§T.6.5), the differential

control that fixes it (§ T.6.6), the depth it requires (§ T.6.7), the ridge geometry it estimates (§ T.6.8), and how clone sizes enter (§ T.6.9).

DEFINITION 2. Let $f_{\text{gen}} = \varphi_{\#} P_{\text{gen}}$ be the density of embedded prototypes and f_{obs} the density of an observed repertoire on embedding space. The *enrichment* is the density ratio $E(z) = f_{\text{obs}}(z)/f_{\text{gen}}(z)$; convergent selection is the region $E(z) > 1$.

NOTATION. A receptor (clonotype) σ is mapped to a point $z = \varphi(\sigma) \in \mathbb{R}^p$ (the prototype embedding, standardised and PCA-projected to $p \approx 65$ coordinates). P_{gen} is the V(D)J generation measure and π_{obs} the sampled-repertoire law; their pushforwards under φ are the densities $f_{\text{gen}} = \varphi_{\#} P_{\text{gen}}$ and $f_{\text{obs}} = \varphi_{\#} \pi_{\text{obs}}$, and $E = f_{\text{obs}}/f_{\text{gen}}$ is their ratio. The selection factor $Q(\sigma) \geq 0$ reweights P_{gen} into π_{obs} (Eq. (5)), with normaliser $Z_Q = \mathbb{E}_{P_{\text{gen}}}[Q]$ and embedding-space average $q(z) = \mathbb{E}[Q \mid \varphi(\sigma) = z]$. We draw N observed and M background points; a ball $B(z, r)$ of radius r contains n_{obs} observed and n_{bg} background points, with target null occupancy $\lambda_0 = \mathbb{E}[n_{\text{obs}} \mid H_0]$; r_1 is the median embedding distance moved by a single CDR3 amino-acid substitution, and a_j (§ T.6.9) the read count of clonotype j . Symbols $\mathbb{E}$ (expectation), Var , $\text{CV} = \sqrt{\text{Var}}/\mathbb{E}$, and Φ^{-1} (standard-normal quantile) are standard.

T.6.1. THE REPERTOIRE AS A MIXTURE, AND WHY THE RAW P_{gen} NULL IS MISCALIBRATED

The observed repertoire is not the generation model. A sampled sequence has survived thymic and peripheral selection, so the observed law is a reweighting of P_{gen} [23, 24],

$$\pi_{\text{obs}}(\sigma) = P_{\text{gen}}(\sigma) \frac{Q(\sigma)}{Z_Q}, \quad Z_Q = \mathbb{E}_{P_{\text{gen}}}[Q], \quad (5)$$

with $Q \geq 0$ the (non-constant) selection factor; the scalar Q of ALICE is its mean-field reduction. Let $q(z) = \mathbb{E}[Q(\sigma) \mid \varphi(\sigma) = z]$ be the local mean selection field.

PROPOSITION 14 (Pushforward ratio is the selection field). $E(z) = q(z)/Z_Q$.

Proof. $f_{\text{obs}}(z) = \int \delta(\varphi(\sigma) - z) P_{\text{gen}}(\sigma) Q(\sigma) / Z_Q d\sigma = q(z) f_{\text{gen}}(z) / Z_Q$. $\square$

So $E \equiv 1$ only if selection is neutral. Two immunological facts make q sharply peaked: *convergent recombination* makes public receptors recur across individuals in proportion to their generation frequency [25, 26, 27], and *thymic selection* is positively correlated with those same generation biases [23], amplifying rather than flattening the hierarchy — a rich-get-richer field concentrated on the convergent, germline-proximal support. The cost is a null that is biased *high* exactly where clones sit.

PROPOSITION 15 (Size-biased bulk fold; the flood). *Averaging the fold over observed clones,*

$$\mathbb{E}_{z \sim f_{\text{obs}}}[E(z)] = \frac{\mathbb{E}_{f_{\text{gen}}}[q^2]}{\mathbb{E}_{f_{\text{gen}}}[q]^2} = 1 + \text{CV}^2(q) \geq 1, \quad (6)$$

strict whenever $\text{Var}(q) > 0$.

Proof. $\mathbb{E}_{f_{\text{obs}}}[E] = \int (q/Z_Q) f_{\text{obs}} = \int (q/Z_Q)^2 f_{\text{gen}} = \mathbb{E}[q^2]/\mathbb{E}[q]^2 = 1 + \text{Var}(q)/\mathbb{E}[q]^2$. $\square$

A test that assumes $E \equiv 1$ therefore rejects across the entire region where q/Z_Q clears the fold threshold. This is measurable: on a VDJDB benchmark in which known antigen-specific ridges are admixed with an unrelated repertoire as noise, a raw P_{gen} background flags 43% of the noise clones as “enriched”, whereas a matched control background (§ T.6.6) flags only 1% — a 46-fold gain in signal-to-noise. The 43% is the selection inflation $1 + \text{CV}^2(q)$, *not* a 43% non-null fraction. The remedy is therefore to recentre the null empirically (§ T.6.5) or to cancel q with a matched control (§ T.6.6). Note q/Z_Q cannot exceed 1 everywhere — both densities integrate to one, so $\mathbb{E}_{f_{\text{gen}}}[q/Z_Q] = 1$ — so it exceeds 1 only on the observed support, which is exactly where clones are tested.

REMARK 1 (Ridges are heterogeneous). The signal is a sum of epitope-indexed ridges, $f_{\text{sig}} = \sum_e w_e g_e$, with $w_e = \sum_{\sigma \in S_e} \pi_{\text{obs}}(\sigma)$ the precursor frequency of epitope e ; the elevation g_e/f_{gen} is *larger* for low- P_{gen} (private) motifs at equal observed count. Inheriting the heavy tail of P_{gen} , the weights w_e span about two orders of magnitude across epitopes and six within one [28]. The polyclonality of a response tracks this precursor frequency: on VDJDB the immunodominant HLA-A*02 influenza epitope GILGFVFTL resolves into 32 distinct convergent mountains, while the CMV A*02 epitope NLVPMVATV gives only 4. No global height threshold separates signal from background across such a range; significance must be assessed locally, and the constructions below are engineered so that it is.

T.6.2. ENRICHMENT AS A DENSITY RATIO: THREE ESTIMATORS AND THEIR RATE

Estimating E as a quotient of two kernel-density estimates is doubly bad: it pays the density-estimation rate twice and blows up where f_{gen} is small. Direct estimation avoids both, and rests on one moment identity [29, 30].

LEMMA 1 (Least-squares characterisation). $\mathbb{E}_{f_{\text{gen}}}[gE] = \mathbb{E}_{f_{\text{obs}}}[g]$ for all $g \in L^2(f_{\text{gen}})$, and $E = \arg \min_g \{\frac{1}{2}\mathbb{E}_{f_{\text{gen}}}[g^2] - \mathbb{E}_{f_{\text{obs}}}[g]\}$.

Proof. $\int g(f_{\text{obs}}/f_{\text{gen}})f_{\text{gen}} = \int g f_{\text{obs}}$; the objective equals $\frac{1}{2}\mathbb{E}_{f_{\text{gen}}}[(g-E)^2]$ up to a constant, minimised uniquely at $g = E$. $\square$

Because the empirical objective uses the two clouds only through sample means, the same target admits three consistent estimators. **(i) Balloon** count-ratio (§ T.6.4), local and carrying an exact test. **(ii) Relative ratio** (RuLSIF) $E_\alpha = f_{\text{obs}}/(\alpha f_{\text{obs}} + (1-\alpha)f_{\text{gen}}) \in [0, 1/\alpha]$, fit in a kernel basis [31]; it is bounded, hence valley-stable, and inverts to $E = (1-\alpha)E_\alpha/(1-\alpha E_\alpha)$, with α the water level — a denominator denser than f_{gen} caps the runaway public-convergence fold at $1/\alpha$. **(iii) Classifier** logit: labelling observed points $y=1$ (prior $\pi = N/(N+M)$) and background $y=0$, a calibrated classifier η satisfies

$$\text{logit } \eta(z) = \log E(z) + \log \frac{\pi}{1-\pi} \quad [32, 33], \quad (7)$$

so any discriminator — featurised by the forward codec φ of Part 2 (`mir/ml/encoder.py`) — recovers $\log E$ up to the known imbalance offset and scales to 10–100M receptors; normalising flows [34] give the two densities explicitly when wanted. The shared offset $\log(N/M)$ is why the control-background and P_{gen} -background detectors estimate the *same* object (§ T.6.3).

PROPOSITION 16 (Direct estimation beats plug-in). *If E is bounded and lies in a class of bracketing entropy $H(\varepsilon) \asymp \varepsilon^{-2\gamma}$, $0 < \gamma < 1$, the regularised least-squares estimator attains $\|\hat{E} - E\|_{L^2(f_{\text{gen}})} = O_p(n^{-1/(2+2\gamma)})$, $n = \min(N, M)$, which is minimax for the likelihood ratio [35]; the plug-in quotient pays the density-estimation rate $n^{-s/(2s+d)}$ per factor plus a small-denominator variance blow-up.*

Proof. By Lemma 1 the population risk of any candidate g is $J(g) = \frac{1}{2}\mathbb{E}_{f_{\text{gen}}}[g - E]^2$ up to an additive constant, so the excess risk of the empirical minimiser $\hat{E}$ over a class $\mathcal{G}$ equals $\frac{1}{2}\|\hat{E} - E\|_{L^2(f_{\text{gen}})}^2$. Empirical-process theory bounds this by an approximation term (the regularisation bias) plus a stochastic term set by the local modulus of the empirical process indexed by $\mathcal{G}$, i.e. by the entropy integral $\int_0^\delta \sqrt{H(\varepsilon)} d\varepsilon \asymp \delta^{1-\gamma}$ when $H(\varepsilon) \asymp \varepsilon^{-2\gamma}$. Choosing the regularisation so that bias and stochastic term are of the same order and solving the resulting fixed point for the critical radius δ gives $\delta \asymp n^{-1/(2+2\gamma)}$; a matching lower bound follows from Fano’s inequality on a δ -packing of $\mathcal{G}$, so the rate is minimax [35]. Boundedness of E keeps the class envelope finite in $L^\infty(f_{\text{gen}})$, which the bound requires. The plug-in quotient $\hat{f}_{\text{obs}}/\hat{f}_{\text{gen}}$ has no such envelope: where $\hat{f}_{\text{gen}} \rightarrow 0$ the ratio’s variance diverges — exactly the instability the bounded relative ratio $E_\alpha \leq 1/\alpha$ of §T.6.2 removes by construction. $\square$

At $p \approx 65$ the nonparametric rate is slow; treat $\hat{E}$ as a locally smoothed, not pointwise-exact, field. The three estimators cross-validate: the classifier gives a fast global E , the balloon supplies calibrated significance, the relative estimator delineates smooth ridge boundaries; disagreement flags estimation error, not signal.

T.6.3. LOCAL SIGNIFICANCE: THE EXACT POINT-PROCESS TEST

Model the two clouds as independent Poisson point processes of intensities $Nf_{\text{obs}}, Mf_{\text{gen}}$ [36]. In a ball $B = B(z, r)$ the counts are Poisson with means $\int_B Nf_{\text{obs}}, \int_B Mf_{\text{gen}}$, independent across disjoint sets (Kingman restriction).

PROPOSITION 17 (Poisson exact test — ALICE). *Under $H_0 : f_{\text{obs}} = f_{\text{gen}}$, with the leave-one-out, Laplace-smoothed null mean*

$$\mu_0 = \frac{(N-1)(n_{\text{bg}}+1)}{M+1}, \quad p = \Pr[\text{Poisson}(\mu_0) \geq n_{\text{obs}}], \quad (8)$$

the enrichment p -value is exact given μ_0 ; for an empty background ball the rule-of-three bound $\mu_0 = 3N/M$ applies. This is (8) as implemented in `mir/density.py`.

Proof. Poisson tail on the restricted process; $(N-1)/(M+1)$ with $+1$ in the numerator is the rule-of-succession (Beta(1, 1)) estimate of the ball probability with self-exclusion, and $3N/M$ is the 95% upper bound on a Poisson mean from zero events. $\square$

PROPOSITION 18 (Conditional binomial — TCRNET). *For independent processes with in-ball means $\mu_{\text{obs}}, \mu_{\text{bg}}$, conditioning on the total $T = n_{\text{obs}} + n_{\text{bg}}$ gives $n_{\text{obs}} \mid T \sim \text{Binomial}(T, \mu_{\text{obs}}/(\mu_{\text{obs}} + \mu_{\text{bg}})) \xrightarrow{H_0} \text{Binomial}(T, N/(N+M))$, the local volume cancelling. This nuisance-free two-sample test [37] is the binomial branch used with a real control background.*

PROPOSITION 19 (ALICE is the rare-event limit of TCRNET). *With $p_{\text{bg}} = (n_{\text{bg}}+1)/(M+1) \sim \lambda_0/M \ll 1$, the Le Cam / Chen–Stein bound [38] gives $d_{\text{TV}}(\text{Binomial}(T, p_{\text{bg}}), \text{Poisson}(Tp_{\text{bg}})) \leq Tp_{\text{bg}}^2$, so the two exact tests agree to $O(\lambda_0 p_{\text{bg}})$.*

Proposition 19 is the theoretical reason a generated P_{gen} background (Poisson) and a real control (binomial) return the *same* enriched set — benchmarked at Jaccard 0.86 against a chance 0.30; ALICE is a special case of TCRNET, and the residual 0.14 is the convergence-field over-call the differential control removes (§T.6.6). One caveat: clonal expansion makes raw counts super-Poisson, so counting must be over *distinct* clonotypes and the null recentred (§T.6.5).

T.6.4. THE BALLOON ESTIMATOR AND THE DETECTION FLOOR

Set the radius $R(z)$ to the distance to the k -th *background* neighbour, $k = \text{round}(\lambda_0 M/N)$. The shared ball volume cancels from the ratio, which is what makes the estimator usable in high dimension [39, 40, 41].

PROPOSITION 20 (Volume-free ratio; homogenised null). *The balloon densities $\hat{f}_{\text{obs}} = n_{\text{obs}}/(NV_p R^p)$, $\hat{f}_{\text{gen}} = n_{\text{bg}}/(MV_p R^p)$ give*

$$\hat{E}(z) = \frac{n_{\text{obs}}/N}{n_{\text{bg}}/M}, \quad k = \text{round}(\lambda_0 M/N) \implies \mathbb{E}[n_{\text{obs}} | H_0] \approx \lambda_0 \text{ for all } z. \quad (9)$$

The null is $\text{Poisson}(\lambda_0)$ independent of the local density $f_{\text{gen}}(z)$; the leading rate is set by the intrinsic dimension $d^ \ll p$ of the receptor manifold, not by p .*

Proof. Fixing k background points fixes the enclosed background mass $\approx k/M$, so the expected observed count is $N \cdot k/M = \lambda_0$; $V_p R^p$ cancels between numerator and denominator, removing the explicit dimension. $\square$

Fixing λ_0 (rather than the radius) is exactly what Remark 1 demands: with precursor frequencies spanning orders of magnitude, a fixed radius would be empty in sparse regions and saturated in dense ones, whereas fixing the null occupancy makes every ball testable on its own scale.

PROPOSITION 21 (Minimum detectable enrichment). *Under H_1 with local fold E , $n_{\text{obs}} \sim \text{Poisson}(E\lambda_0)$, and a one-sided level- q test is cleared at 50% power once*

$$E^*(q, \lambda_0) = 1 + \frac{c(q)}{\sqrt{\lambda_0}}, \quad c(q) = \Phi^{-1}(1 - q); \quad (10)$$

under Benjamini–Hochberg over N tests $c \approx \sqrt{2 \ln N}$. Thus E^ falls only as $\lambda_0^{-1/2}$ while the radius grows as λ_0^{1/d^*} (essentially flat in high d^*) and must stay $\leq r_1$, the median one-substitution embedding drift, to resolve a motif ridge; balancing these gives the default $\lambda_0 = 3$ (detectable ~ 2.3 -fold) with the range $[1, 5]$.*

Proof. By Proposition 20 the null count is $n_{\text{obs}} \sim \text{Poisson}(\lambda_0)$, of mean and variance λ_0 . The one-sided level- q test rejects at $n_{\text{obs}} \geq \tau_q$ with $\tau_q = \min\{k : \Pr[\text{Poisson}(\lambda_0) \geq k] \leq q\}$, and the normal approximation gives $\tau_q \approx \lambda_0 + c(q)\sqrt{\lambda_0}$, $c(q) = \Phi^{-1}(1 - q)$. Under H_1 the count has mean $E\lambda_0$,

so the test attains 50% power once $E\lambda_0 \geq \tau_q$, i.e. once $E \geq 1 + c(q)/\sqrt{\lambda_0}$ — Eq. (10). Demanding power $1 - \beta$ replaces $c(q)$ by $c(q) + \Phi^{-1}(1 - \beta)$, and under Benjamini–Hochberg the effective per-test level tightens to $\approx q/N$, whence $c \approx \Phi^{-1}(1 - q/N) \approx \sqrt{2 \ln N}$. The exact Poisson tail at $q=0.05$ has critical counts $\tau = 4, 7, 10$ for $\lambda_0 = 1, 3, 5$, giving $E^* = \tau/\lambda_0 \approx 4.0, 2.3, 2.0$. Finally a ball holding λ_0 of the M background points encloses mass $\lambda_0/M \propto R^{d^*}$ on the intrinsic-dimension- d^* manifold, so $R \propto \lambda_0^{1/d^*}$ — nearly flat for large d^* , so raising λ_0 costs almost no resolution until R reaches the motif width r_1 , which pins the operating point at $\lambda_0 = 3$. $\square$

Pinning $R \approx r_1$ makes the ball the continuous analogue of the Hamming-1 shell, and $E(z)$ is its $r \rightarrow 0$ limit, $E(z) = \lim_{r \rightarrow 0} [n_{\text{obs}}(B_r)/(N-1)]/[n_{\text{bg}}(B_r)/M]$. This explains the measured continuous–discrete concordance: the Spearman correlation between the balloon count and the discrete Hamming-1 count is $\rho = 0.37, 0.34, 0.11, -0.05$ at $r = 0.5, 1, 2, 3 \times r_1$ — high at one substitution (where the ball coincides with the graph neighbourhood), decaying past it as the ball integrates several mode basins. The continuous test *generalises* the graph test rather than reproducing it.

T.6.5. EMPIRICAL-NULL CALIBRATION: THE WATER LEVEL

By Proposition 15 the theoretical P_{gen} null is wrong by a global factor; before any multiplicity control it must be recentred to the data. This is Efron’s empirical null [42], and it precedes FDR.

PROPOSITION 22 (Enrichment is an inverse local FDR). *In the two-groups model $f = \pi_0 f_0 + \pi_1 f_1$ with a calibrated null f_0 , the local false-discovery rate is $\text{fdr}(z) = \pi_0 f_0(z)/f(z) = \pi_0/E(z)$; declaring $E(z) \geq 1/\alpha$ is a conservative $\text{fdr} \leq \alpha$ rule [42]. With the raw null $f_0 = f_{\text{gen}}$ one instead has $\text{fdr} = \pi_0 \mu_0/E$ — off by the miscalibration of (6).*

PROPOSITION 23 (Water level = one-sided empirical-null recentring). *Let the true background be $f_{\text{bg}}(z) = s f_{\text{gen}}(z)$ on the tested support with unknown inflation $s > 1$. If the ridge fraction is below $\frac{1}{2}$, then*

$$c = \max\left(\frac{\text{median}_i n_{\text{obs},i}}{\text{median}_i \mu_{0,i}}, 1\right), \quad \mu_0 \leftarrow c \mu_0 \implies \text{median}_i \text{fold}_i \approx 1, \quad (11)$$

and c is a robust (breakdown $\frac{1}{2}$) consistent estimate of s .

Proof. Write $\log \hat{E} = \log s + \varepsilon$ with $\varepsilon \approx 0$ on the bulk and > 0 on ridges; if ridges occupy under half the tests, $\text{median}(\log \hat{E}) = \log s$, so (11) estimates s and $\max(\cdot, 1)$ enforces one-sidedness (never deflate below the P_{gen} floor). This is Efron’s empirical-null location correction [42], equivalently the median-of-ratios size factor of DESeq [43]. Being a median it is bounded above by the mean $1 + \text{CV}^2(q)$ of (6). $\square$

This is the `calibrate="median"` step of `mir/density.py`: because the signal is $< 5\%$ in a naive repertoire it barely moves the median, so recentring the bulk to $\text{fold} \approx 1$ is robust. After recentring, overlapping balls are positively associated, so Benjamini–Hochberg controls the FDR under positive-regression dependence [44, 45]. The failure mode is sharp: a single scalar c removes only a *constant* inflation. A spatially varying selection field $s(z)$ — or a tetramer-sorted sample with ridge fraction $\geq \frac{1}{2}$ — breaks the median, and the only fix is a real control.

T.6.6. THE DIFFERENTIAL CONTROL: CANCELLING CONVERGENCE POINTWISE

PROPOSITION 24 (Shared-field cancellation). *Let case and control share the selection/convergence field, $f_{\text{case}} = (q/Z_Q)f_{\text{gen}}(1 + a_1)$ and $f_{\text{ctrl}} = (q/Z_Q)f_{\text{gen}}(1 + a_0)$, with $a_i \geq 0$ the condition-specific antigen expansion. Then*

$$R(z) = \frac{f_{\text{case}}(z)}{f_{\text{ctrl}}(z)} = \frac{1 + a_1(z)}{1 + a_0(z)} \perp q, Z_Q, f_{\text{gen}}, \quad (12)$$

independent of the shared field; under H_0 ($a_1 \equiv a_0$) $R \equiv 1$ with no water-level offset, even where $q(z)$ varies spatially. The binomial test of Proposition 18 is its finite-sample plug-in; ALICE is the special case control $\equiv f_{\text{gen}}$ after recentring.

Proof. Direct division cancels the shared factor $(q/Z_Q)f_{\text{gen}}$; in the small-ball limit the case/control odds converge to $(N_{\text{case}}/N_{\text{ctrl}})R(z)$, a Radon–Nikodym change of reference from the generation model to the control. $\square$

PROPOSITION 25 (Control-absent motifs are maximally identifiable). *If the control lacks a motif on a region A ($f_{\text{ctrl}} \rightarrow 0$, $f_{\text{sig}} > 0$) then $R \rightarrow \infty$ on A (odds-ratio $\rightarrow$ relative-risk regime [46]); under the pure-control anchor $\text{ess inf}_z(f_{\text{sig}}/f_{\text{ctrl}}) = 0$ the mixing weight is identified as $1 - \pi = \text{ess inf}_z R(z)$ [47].*

This is why the recommended course of action prefers a biological control (pre- vs post-vaccination, patient vs healthy, B27 $\pm$) over the raw P_{gen} null, and it is what the benchmarks show. The public CASSVGL[YF]STDTQYF (TRBV9/TRBJ2-3) motif is control-absent in ankylosing spondylitis, so $R \rightarrow \infty$ on its ridge: it surfaces among enriched B27+ synovial hits (9) and is absent in B27– (0). In yellow-fever vaccination the shared naive field cancels, isolating the expansion: day-15 versus day-0 enriched hits matching the LLW/A*02 reference are 63 > 35 (of 129 vs 79 distinct such clonotypes present). Cancellation requires case and control to share q — match HLA, depth, and cell subset; a contaminated control attenuates R toward 1, making the differential test conservative, not anti-conservative.

T.6.7. SAMPLING DEPTH: WHY THE FULL REPERTOIRE

Subsampling a repertoire with retention probability s is Poisson thinning: counts scale to Poisson($s\mu$) [36].

PROPOSITION 26 (Thinning cost and detection boundary). *With signal mass Λ_e and null mean λ_0 , the test’s non-centrality $\delta = \Lambda_e/\sqrt{\lambda_0}$ thins as $\delta_s = \sqrt{s}\delta$. Writing depth $D = sN$ and $\lambda_0 \propto Dw_e$, the minimum detectable precursor frequency is*

$$w_e^{\min}(D) = \frac{c(\alpha, \beta, \lambda_0)}{D}, \quad (13)$$

and a clone is even observed only with probability $1 - e^{-Dw_e}$ [48].

Proof. Poisson thinning with retention probability s maps the intensity of every region from μ to $s\mu$ [36]. At depth $D = sN$ a ridge of precursor frequency w_e contributes an expected in-ball signal count

$\propto Dw_e$ against a null of mean λ_0 and standard deviation $\sqrt{\lambda_0}$, so by the Gaussian approximation of Proposition 21 its power is $\Phi(Dw_e/\sqrt{\lambda_0} - \Phi^{-1}(1-\alpha))$, reaching $1-\beta$ exactly when $Dw_e \geq (\Phi^{-1}(1-\alpha) + \Phi^{-1}(1-\beta))\sqrt{\lambda_0}$; that is $w_e \geq w_e^{\min}(D) = c/D$ with $c = (\Phi^{-1}(1-\alpha) + \Phi^{-1}(1-\beta))\sqrt{\lambda_0}$, which is Eq. (13). Independently, the clone is *present* at all only if its count $\text{Poisson}(Dw_e)$ is nonzero, an event of probability $1 - e^{-Dw_e}$ [48]. $\square$

Antigen clusters are rare in *absolute* number, so subsampling by s inflates the detection floor by $1/s$ and drops sparse ridges below it — and at reduced depth low- w_e clusters are often not sampled at all. Hence the full repertoire must be processed; save memory with junction-only embedding or a PCA-fit cap, never by subsampling. Because w_e is heavy-tailed (Remark 1), the boundary $Dw_e = c$ is crossed at a different depth for every epitope — one more reason significance is local, never a global height.

T.6.8. RIDGE DELINEATION AND PER-RIDGE TESTS

The antigen-specific groups are the high components of E : its *density ridges* [49] and upper-level-set components [50, 51].

PROPOSITION 27 (Ridge extraction as level-set estimation). *The connected components of $\{z : E(z) > \tau\}$ form the density cluster tree; distinct epitope motifs are distinct components appearing at different levels, so no single global τ certifies all of them, but each is certified locally. The FDR-significant hit set $\{q < \alpha, \text{fold} > 1\}$ estimates one level set, and DBSCAN with $\varepsilon = r_1$, $\text{minPts} \in \{2, 3\}$ links its points into motifs — its core-point rule is itself a superlevel-set threshold [52, 51], so clustering and the Poisson test stay mutually consistent; $\varepsilon \gg r_1$ over-merges, the same regime where the continuous–discrete concordance collapses.*

Each motif is then labelled: a two-sample Poisson test versus a matched control for a condition, and, for an epitope, the binomial test of [22] — successes = receptors in the ridge matching the epitope, trials = ridge size, probability = the repertoire-wide annotation rate — with Benjamini–Hochberg across ridges. Because each ridge is judged against its own local background, the height heterogeneity of Remark 1 is handled automatically; background subtraction removes the smooth selection field and the surviving convergent groups are the antigen-specific clusters.

T.6.9. CLONAL ABUNDANCE: SCORING WITH CLONE SIZES, ROBUSTLY

The tests so far count *distinct* clonotypes, discarding each clone’s read count a_j . This is deliberate — read counts are heavy-tailed (roughly Zipf or log-normal), so a single hyperexpanded clone would dominate a raw read-weighted count — but it throws away real signal. Two independent signatures mark an antigen-driven response: *breadth*, a group of similar sequences produced by convergent diversification (measured by the neighbour count n_{obs}), and *depth*, a clonotype present at high copy number through clonal expansion (measured by a_j). A textbook antigen-specific cluster shows both, but the corners matter: a convergent cluster of singletons has breadth without depth, and a hyperexpanded *orphan* — large a_j , few or no neighbours, the case [22] recommends inspecting individually

— has depth without breadth. A complete detector reads both channels, and uses size to boost the score without letting its tail rule the outcome.

A VARIANCE-STABILISED WEIGHTED COUNT. Replace the in-ball count by a weighted mass

$$S(z) = \sum_{j \in B(z,r)} g(a_j), \quad (14)$$

with g non-decreasing and *concave* — for instance $g(a) = \log(1 + a)$ or the Anscombe root $g(a) = \sqrt{a + \frac{3}{8}}$. Concavity is precisely robustness: the leverage of clone j on S is the marginal $g'(a_j)$, which *decreases* with size, so a size-1000 clone amid size-1 neighbours contributes a bounded increment, not a thousandfold one. The endpoints are the read count $g(a) = a$ (constant leverage, Zipf-dominated) and the shipped distinct count $g(a) = 1$ (zero leverage, no size information); g is a tunable sensitivity–robustness dial between them.

PROPOSITION 28 (Compound-Poisson null for the weighted count). *Under H_0 the in-ball clones are a $\text{Poisson}(\lambda_0)$ -sized sample with independent abundances $A \sim \nu$ (the null size law), so*

$$\mathbb{E}[S \mid H_0] = \lambda_0 \mathbb{E}_\nu[g(A)], \quad \text{Var}[S \mid H_0] = \lambda_0 \mathbb{E}_\nu[g(A)^2], \quad (15)$$

with dispersion index $\text{Var}/\mathbb{E} = \mathbb{E}_\nu[g(A)^2]/\mathbb{E}_\nu[g(A)]$, equal to 1 for $g \equiv 1$ (a Poisson null) and $\gg 1$ for $g(a) = a$ under a heavy-tailed ν . A negative-binomial (Gamma–Poisson) tail matched to these two moments gives a calibrated one-sided p -value; a variance-stabilising g keeps the null close to Poisson, so the exact test of §T.6.3 still applies.

Proof. Write $S = \sum_{i=1}^K g(A_i)$ with $K \sim \text{Poisson}(\lambda_0)$ and $A_i \sim \nu$ i.i.d. and independent of K . Wald’s identity gives $\mathbb{E}[S] = \mathbb{E}[K] \mathbb{E}[g(A)]$, and the law of total variance gives $\text{Var}[S] = \mathbb{E}[K] \text{Var}[g(A)] + \text{Var}[K] \mathbb{E}[g(A)]^2 = \lambda_0 (\text{Var}[g(A)] + \mathbb{E}[g(A)]^2) = \lambda_0 \mathbb{E}[g(A)^2]$, using $\mathbb{E}[K] = \text{Var}[K] = \lambda_0$. The dispersion index follows, and for $g \equiv 1$ it is 1, recovering the $\text{Poisson}(\lambda_0)$ null of Proposition 20. $\square$

THE ORPHAN CHANNEL. A hyperexpanded clone with few neighbours is invisible to S yet significant on its own. Under neutral sampling at depth D its size is $\text{Poisson}(Df_j)$ with f_j its frequency, so its abundance p -value against the null size law,

$$p_j^{\text{size}} = \Pr_{A \sim \nu}[A \geq a_j], \quad (16)$$

tests expansion independently of any neighbourhood. Combining the breadth and depth channels by Fisher’s rule, $-2(\log p_j^{\text{nbr}} + \log p_j^{\text{size}}) \sim \chi_4^2$ under the (approximately independent) null, recovers the orphan when $p^{\text{nbr}} \approx 1$ but p^{size} is small, and multiplicatively sharpens a cluster that is both convergent and expanded.

REMARK 2 (Rarefaction sets the size law). The null size law ν and the breadth/depth balance are read from the frequency-of-frequencies spectrum — the counts $f_1, f_2, \dots$ of singletons, doubletons, and larger clones [48]. The Good–Turing mass on unseen clones is $\approx f_1/D$, and the singleton fraction is a saturation gauge: a repertoire heavy in singletons is undersampled, so breadth carries the signal, whereas a saturated repertoire lets depth dominate. Singletons and doubletons are the convergence substrate; the 3+ and hyperexpanded classes are where the weight g and the orphan test earn their keep.

This abundance layer leaves the geometry untouched: S replaces n_{obs} in the same balloon, the same recentring (§ T.6.5) and FDR apply, and the orphan test is a per-clonotype side-channel. Clone sizes then boost the score exactly where expansion co-occurs with convergence, while the concavity of g keeps the Zipf tail from ruling the result. The shipped `mir/density.py` uses $g \equiv 1$; the weighted and orphan channels are the abundance-aware extension.

T.6.10. COURSE OF ACTION

The theory fixes every parameter, giving a graph-free reincarnation of TCRNET/ALICE (`mir/density.py`):

1. **Background.** With a matched control, use it and the binomial test (Prop. 24); otherwise generate $M \geq 5N$ samples from P_{gen} and use the Poisson test. Prefer a control whenever one exists.
2. **One basis.** Embed both clouds through one prototype set and one PCA rotation (Lemma 1); $E(z)$ is meaningless across mismatched bases.
3. **Adaptive ball.** Per observed z , radius = distance to the k -th background neighbour, $k = \text{round}(\lambda_0 M/N)$, $\lambda_0 = 3$ (Prop. 20, 21); count $n_{\text{obs}}, n_{\text{bg}}$; form μ_0 by (8).
4. **Recentre** the P_{gen} null by (11) (Prop. 23); a differential control needs none.
5. **Test** the exact upper tail (Poisson or binomial), Benjamini–Hochberg $q < 0.05$, fold > 1 ; the detection floor is $E^* \approx 1 + c(q)/\sqrt{\lambda_0}$.
6. **(Optional) clone sizes.** Swap n_{obs} for the variance-stabilised weighted count $S(z) = \sum g(a_j)$ and add the per-clonotype orphan test (§ T.6.9, Prop. 28) to score expansion alongside convergence.
7. **Cluster** hits by DBSCAN, $\varepsilon = r_1$ (Prop. 27), and test each motif for label enrichment. Process the full repertoire (Prop. 26).

Every default traces to a theorem rather than to tuning; the same estimand $E = f_{\text{obs}}/f_{\text{gen}}$ underlies the graph tests it replaces, the neural classifier of Part 2, and the smooth relative-ratio field.

T.7. PRIVACY: THE EMBEDDING AS A KEYED, LOSSY REPRESENTATION

Immune repertoires are individually identifying — like genomic data, a person’s TCRs are a stable, near-unique fingerprint — so whether an embedding may be shared in place of raw sequences is a practical question with a clean geometric answer. Reproducing a coordinate requires two secrets: the prototype set $P = \{p_k\}$ and the PCA rotation W fitted to the cohort (together with the centring μ and scaling). Write the released representation as $Z = (\varphi(X) - \mu) W \in \mathbb{R}^{n \times m}$ with $m \ll 3K$.

THREE BARRIERS TO RECONSTRUCTION. (i) *Lossy projection.* PCA to m components discards a $(3K - m)$ -dimensional subspace, so $\varphi \mapsto Z$ is many-to-one and non-invertible on the discarded directions *even given* W — an information-theoretic barrier no decoder can cross. (ii) *Unknown anchors.* Recovering x from $\varphi(x) = (d(x, p_k))_k$ is reconstruction from distances to *unknown* reference points in a discrete sequence space, a hard combinatorial inverse problem without P . (iii) *Data-dependent mixing.* W is estimated from the cohort, not a public transform; without it the released

numbers are linear combinations of unknown prototype distances. Releasing Z alone is therefore a **keyed, lossy pseudonymisation**: the pair (P, W) is a secret key, and without it reconstruction reduces to a hard inverse problem stacked on an irreversible compression.

WHAT IT DOES NOT PROTECT. The embedding is built to preserve pairwise geometry (Prop. 1–3), and that is precisely what a linkage adversary exploits. A key-holder — or anyone able to embed a reference sequence under the same (P, W) — can (a) test membership by embedding a candidate and matching it against Z (re-identification / membership inference), (b) link the same patient across releases through preserved distances, and (c) invert approximately with a trained decoder: the Part 2 inverse codec (`mir/ml/decoder.py`) demonstrates that φ is approximately invertible *given the key*. Distance preservation is thus double-edged — the source of the embedding’s analytic value and of its linkage risk. Keyed pseudonymisation is not anonymisation.

A DIFFERENTIAL-PRIVACY MECHANISM THE GEOMETRY HANDS US. The Lipschitz property of Prop. 1 turns the embedding into a clean substrate for a formal guarantee. Replacing one receptor moves its coordinate by $\|\varphi(x) - \varphi(x')\|_2 \leq \sqrt{K} d(x, x') \leq \sqrt{K} d_{\max}$, so $X \mapsto \varphi(X)$ has bounded ℓ^2 sensitivity.

PROPOSITION 29 (Private release of the embedding). *Releasing $\tilde{Z} = (\varphi(X) - \mu)W + N$ with N Gaussian of scale $\sigma \propto \sqrt{K} d_{\max} \|W\|_2 / \varepsilon$ satisfies record-level (ε, δ) -differential privacy, and by Prop. 1 the induced distortion of any pairwise embedding distance D_{ij} is $O(\sigma\sqrt{m})$ — a privacy/utility trade-off governed entirely by the sensitivity constant $\sqrt{K} d_{\max}$.*

Proof. Take two repertoires differing in a single record, x_i versus x'_i . Only row i of $\varphi(X)$ changes, and by the 1-Lipschitz coordinates of Prop. 1, $\|\varphi(x_i) - \varphi(x'_i)\|_2 \leq \sqrt{K} d(x_i, x'_i) \leq \sqrt{K} d_{\max}$. Composing with W , the released row moves by $\|(\varphi(x_i) - \varphi(x'_i))W\|_2 \leq \sqrt{K} d_{\max} \|W\|_2 =: \Delta_2$, the record-level ℓ^2 -sensitivity. The Gaussian mechanism, adding $N \sim \mathcal{N}(0, \sigma^2 I)$, is a standard (ε, δ) -differentially private release whenever $\sigma \geq \Delta_2 \sqrt{2 \ln(1.25/\delta)} / \varepsilon$, i.e. for $\sigma \propto \sqrt{K} d_{\max} \|W\|_2 / \varepsilon$. For utility, the perturbed distance $\tilde{D}_{ij} = \|(z_i - z_j) + (N_i - N_j)\|$ obeys $|\tilde{D}_{ij} - D_{ij}| \leq \|N_i - N_j\|$ by the triangle inequality, and $N_i - N_j \sim \mathcal{N}(0, 2\sigma^2 I_m)$ has $\mathbb{E}\|N_i - N_j\|^2 = 2m\sigma^2$, so the distortion is $O(\sigma\sqrt{m})$. $\square$

RECOMMENDATION. For patient data, treat (P, W) as a managed secret, release only the truncated projection Z (never φ or the invertible residual), and — when a provable guarantee is required rather than obscurity — add the calibrated noise of Prop. 29, whose cost is a quantifiable, Lipschitz-bounded distortion of the very distances the analysis depends on.

T.8. INSIGHTS, REPRODUCTION, AND OPEN PROBLEMS

A UNIFYING VIEW. The results above place TCREMP in a well-understood corner of metric geometry. The map φ is the empirical (Nyström) feature map of the alignment kernel; its ℓ^∞ read-out is the Kuratowski isometry, its ℓ^2 read-out is Landmark MDS, and its PCA is the classical scaling of the landmark columns. Three observations may be of independent interest beyond immunology: (a) taking the landmark measure to be a *generative model* of the data (P_{gen}) rather than a random subsample makes the embedding a sketch of each point’s relation to the data-generating process, and makes prototype choice provably robust (Prop. 7); (b) high-dimensional prototype-embedding

distances obey a generalized extreme value law (Prop. 8), a clean consequence of distance concentration that is easy to mistake for Gaussian; and (c) the germline rank bound (Prop. 12) shows how categorical side-information injects exactly-low-rank structure into an otherwise full-rank distance embedding.

REPRODUCTION. Every quantitative statement is regenerated by `mir/bench/theory.py` and the data/figure scripts in this directory (`gen_theory_data.py`, the `*.gp` gnuplot scripts, `diagram_embedding.dot`); Table 1 collects the numbers.

Table 1: Theory predictions and their v3 reproduction (`mir/bench/theory.py`; $n = 2500$ CDR3 β). Paper values from [1], suppl. S1–S3.

Claim	Paper	mirpy (Smith–Waterman)	mirpy (gapblock v3)
S2 / Prop. 3: $\text{corr}(D_{ij}, d_{ij})$	0.56	0.64	0.49
S1 / Prop. 8: law of d_{ij}	$\text{Gamma} > \text{Normal}$	$\text{Gamma} > \text{Normal}$	$\text{Gamma} > \text{Normal}$
S1 / Prop. 8: law of D_{ij}	GEV, $\xi = +0.11$	GEV $\gg$ Normal, $\xi \approx 0$	GEV $\gg$ Normal, $\xi \approx 0$
S3 / Prop. 7: real vs. model prototypes	0.96	—	0.97

OPEN PROBLEMS. Four questions are left open. (1) Sharp distortion constants for the lower bound complementary to Prop. 1 when prototypes are P_{gen} -sampled, i.e. a Bourgain-type [53] guarantee tuned to the recombination measure rather than to the worst case. (2) The exact extreme-value index ξ of Prop. 8 as a function of the substitution matrix and prototype count. (3) A mutation-corrected metric for IGH that makes Prop. 7 hold under SHM (§T.5). (4) Consistency and rates for the three density-ratio estimators of §T.6 (balloon, relative [31], and classifier-based [33]) on the PCA-projected embedding, together with a per-epitope detection limit in the sense of Prop. 26 as a function of precursor frequency and motif generation probability, and calibration of the neural classifier’s significance — which would put continuous background subtraction on the same footing as the graph tests it replaces.

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